

The Impact of Class Struggle on the Capitalist Long Wave

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After publishing "Understanding the Capitalist Long Wave in Depth" I began to think about the impact of class struggle on the long wave cycle. For a view long current in left-wing academic and intellectual circles holds that the proletariat's class struggle lowers the rate of exploitation, thereby hastening the fall of the rate of profit and bringing the upswing to an end more quickly. This is the profit squeeze theory. Although I hold to the theory that a rising organic composition of capital causes the rate of profit to fall, I do not wholly reject the profit squeeze theory; in fact I treat the profit squeeze as a supplementary mechanism. Recently a reader left a comment asking about the role of class struggle, and I felt it deserved serious treatment. In this article I shall introduce a class struggle variable into the equations and derive its effect on the cycle.

Class struggle affects the rate of surplus value, the rate of surplus value affects the rate of profit, and profit is the purpose of capitalist production. From a Marxist standpoint, therefore, it stands to reason that class struggle affects the capitalist economic cycle. In most eras, however, the level of the proletariat's class struggle is basically determined by the size of the industrial reserve army. If the industrial reserve army keeps growing, the proletariat's class struggle is weak, and even when it struggles it is on the defensive, resisting further attacks by the bourgeoisie. But if the industrial reserve army is exhausted, the proletariat's struggle grows stronger. This is also how the authors of *Economic Growth and Long Cycles* handle class struggle.

According to Marx (*Capital*, Volume I, Chapter 25), the growth rate of the rate of surplus value g_e is inversely proportional to changes in the labour force (or, equivalently, investment in variable capital s_V). Glombowski (1983), for instance, assumed $g_e = \varepsilon_0 - \varepsilon_1 s_V$, where ε_0 and ε_1 are small positive constants. In this study we set a relatively simple relationship, namely $g_e = -s_V$. —
Economic Growth and Long Cycles, 8.2.1

But in certain eras, when the proletariat's economic struggle has already combined with political struggle and the proletariat has already brought about considerable change in the politics of the bourgeois state, changes in the rate of surplus value are subject not only to the influence of the industrial reserve army but also directly to the influence of trade unions, communist parties and other left-wing parties. In such a scenario of class struggle going beyond ordinary economic fluctuation, what should the change in the cycle be?

I attempt to introduce a new parameter CS, standing for class struggle. If CS is 0, the situation is the same as that assumed in Economic Growth and Long Cycles. If CS is negative, it means that for some reason the proletariat's level of class struggle is exceptionally weak, while a positive CS indicates that political and ideological factors have raised class struggle to a higher level.

If I change $g_e = -s_V$ to $g_e = -sV - CS$, then the first equation, $r' = -a_1 \cdot (sC - sV) \cdot r^2 + a_2 \cdot (\delta + \tau - sV) \cdot r$, becomes $r' = -a_1 \cdot (sC - sV) \cdot r^2 + a_2 \cdot (\delta + \tau - sV - CS) \cdot r$.

Substituting different values of CS and solving the system of equations, we observe that CS has an inverse effect on the cycle: the larger CS is, the longer the cycle, and the greater the share of the upswing within it. In other words, the proletariat's class struggle lengthens the upswing and shortens the downswing:

Table 1 — Cycle statistics of sC and sV and profitability over the CS grid.

CS	t_dom (sV<0)	t_dom (share>1)	T_lin (eigen.)	sC cycles (y)	sC T_med (y)	sC rise (y)	sC fall (y)	sC rise share	sV T_med (y)	sV rise share	r_mean [0,120]	r_end
-0.100	126	87	41.7	[42.2, 45.0]	43.6	23.7	19.9	54%	44.4	50%	0.1498	0.1146
-0.050	129	87	41.7	[42.4, 45.8]	44.1	24.4	19.7	55%	45.1	50%	0.1454	0.1050
-0.020	130	87	41.7	[42.6, 46.3]	44.4	24.9	19.5	56%	45.5	51%	0.1428	0.0993
-0.010	131	87	41.7	[42.6, 46.4]	44.5	25.1	19.5	56%	45.6	51%	0.1419	0.0975
-0.005	131	86	41.7	[42.6, 46.5]	44.6	25.1	19.4	56%	45.7	51%	0.1415	0.0966
0	131	86	41.7	[42.7, 46.6]	44.6	25.2	19.4	57%	45.8	51%	0.1411	0.0956
+0.005	132	86	41.8	[42.7, 46.7]	44.7	25.3	19.4	57%	45.8	51%	0.1407	0.0947
+0.010	132	86	41.8	[42.7, 46.8]	44.7	25.4	19.3	57%	45.9	51%	0.1403	0.0938
+0.020	133	86	41.8	[42.8, 46.9]	44.8	25.6	19.3	57%	46.1	51%	0.1394	0.0920
+0.050	135	86	41.8	[42.9, 47.4]	45.2	26.1	19.1	58%	46.5	51%	0.1370	0.0865
+0.100	202	85	41.8	[43.2, 48.4, 56.4]	48.4	28.3	20.1	59%	50.1	52%	0.1331	0.0416

Source: https://github.com/sacanlee/long_wave_ode/blob/main/docs/class_struggle_report.md#class-struggle-in-the-long-wave-model-the-closure-g_e--sv-and-an-extension-with-an-autonomous-wage-push-term-cs

This conclusion is highly counter-intuitive. But looked at from another angle it is understandable. The cause of capitalist crises is over-accumulation. The higher the rate of surplus value, the faster accumulation proceeds and the more severe over-accumulation becomes. The proletariat's class struggle slows capital accumulation, lengthening the boom while reducing the severity of over-accumulation, lessening the depth of the crisis and shortening the depression.

The long wave cycles of history seem to bear this out. The longest upswing in history was the post-war golden age from 1940 to 1966. That period saw the highest level of proletarian struggle, organisation and political consciousness in capitalist history, and its upswing lasted a full 26 years. By comparison, the upswings of the nineteenth century lasted only 22 years (1850–1872) and 17 years (1896–1913, which might have stretched to about 20 had there been no First World War), while the most recent long wave upswing lasted only 18 years (1982–2000). Since conditions in the nineteenth and twentieth centuries differed considerably, the most telling comparison is between the fourth and fifth cycles. The fifth cycle began with the neoliberal revolution; in both the centre and the periphery the population lurched sharply to the right, proletarian struggle fell to a historic low, capital accumulation accelerated, the upswing shortened, crises became violent and depressions lengthened. On this point, see the section "Historical correspondence of the CS levels" in the document on GitHub, Class Struggle in the Long-Wave Model: The Closure $g_e \cong -sV$ and an Extension with an Autonomous Wage-Push Term (CS).

If this adjustment to the model and the related derivations are correct, then the contradiction between reform and revolution is in effect dissolved. The proletariat's struggle for reforms, in whatever form, is in essence aimed at lowering the rate of surplus value, that is, at raising CS and lowering g_e . If the proletariat's reformist struggle does not hasten the coming of crisis but instead postpones its outbreak and reduces its intensity, then the proletariat's economic struggle is not only the "school" and "training ground" of revolutionary political struggle but also has a real economic significance. Moreover, proletarian struggle lowers the rate of accumulation, thereby slowing the pace at which capital builds up the industrial reserve army during the boom. Politically, this means that although the proletariat cannot shake off its illusions about reformist politics such as trade unions, it is not easily ensnared by illusions about the capitalist free market.

Generally speaking, the proletariat's strength declines during the downswing, so CS finds it hard to remain positive. But if, for some reason, CS does remain positive during the downswing — that is, if the recession is unable to weaken the proletariat's militancy automatically and does not dissolve proletarian politics — then two situations arise:

- 1) If CS remains positive and exceeds a certain threshold (in this example the threshold is $CS \gtrsim +0.115$), the rate of profit cannot recover but continues downwards, approaching zero, and the capitalist order falls into a deadlock of class struggle.

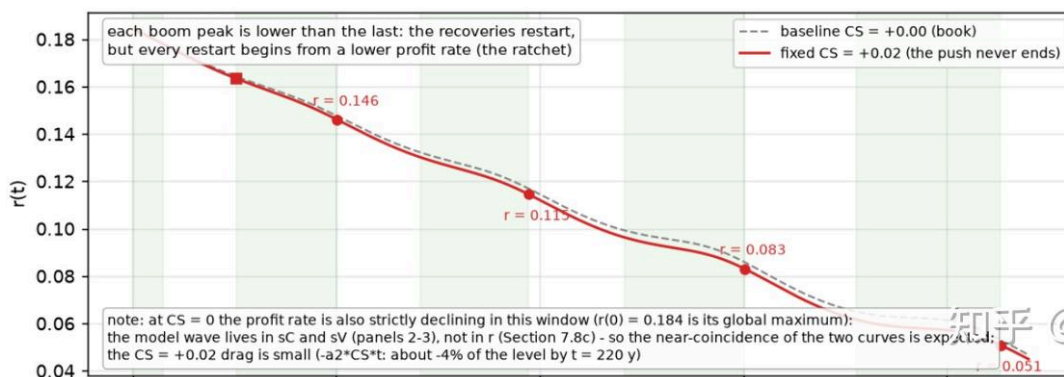
Oscillation death at high CS and the domain boundary (book parameters, $n = 0.015$, 300-year horizon)
 - 0.115 the wage share is never purged and r slides to $\sim 1e-3$ - $1e-2$; above CS ~ 0.35 - 0.5 the oscillation itself dies - and the shares
 & the premise $s_C + s_V \leq 1$, so these high-CS runs are extrapolated regime statements rather than economically meaningful traj

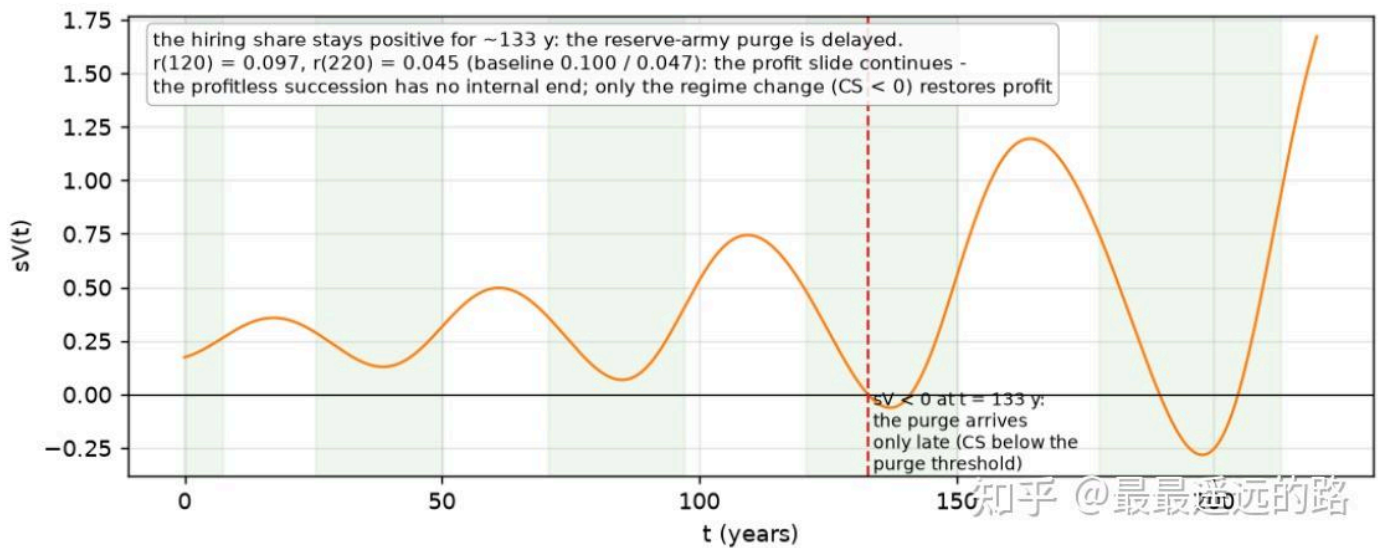


Source: https://github.com/sacanlee/long_wave_ode/blob/main/figures/class_struggle_oscillation_death.png

2) If CS is positive but not especially large (0.02 in this example), below the threshold, then fluctuations in the rate of profit occur but it keeps falling, spiralling downwards. In this scenario "economic recovery" has little practical meaning. As profit shrinks, accumulation slows and total new hiring falls, so unemployment rises. That is to say, workers already in post cannot be laid off and their wages cannot be pushed down, while new entrants to the labour force and those already unemployed cannot find work. The larger CS is — the fiercer the struggle — the more serious long-term unemployment becomes.

Pointless recoveries at a fixed wage push: CS = +0.02 held constant over the whole run (book parameters, $n = 0.015$) the accumulation wave keeps cycling while the profit rate ratchets down - model time, not calibrated to calendar years





Source: https://github.com/sacantee/long_wave_ode/blob/main/figures/class_struggle_ratchet_recoveries.png

In these circumstances, in order to restore the capitalist order the bourgeoisie must launch a class struggle aimed at smashing the proletariat's organisations. And the proletariat then faces a choice: either disarm and accept the bourgeoisie's adjustments, or abolish private property through revolution, abolish capital, and establish a planned economy based on public ownership.

A note:

This article introduces the CS parameter in order to illustrate the impact of class struggle. But in an actual capitalist society CS should not be a parameter but a variable, itself subject to the influence of the economic cycle. The conclusions of this article can therefore only be suggestive, and are meant to show that class struggle does have an impact on the long wave cycle, and that this impact is complex and many-sided.

References:

GitHub project: https://github.com/sacantee/long_wave_ode

Summary

document:

https://github.com/sacantee/long_wave_ode/blob/main/docs/class_struggle_report.md

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